

9. Conclusion

A complete digital communication model for the message "ROY" was implemented in MATLAB. The text was converted into 8-bit ASCII data, transmitted as an LSB-first synchronous serial stream and represented by a 1 kbps unipolar NRZ signal. The data was then modulated using BASK, 4-ASK, 16-ASK, QPSK, 8-PSK and 16-QAM and transmitted through an AWGN channel. The receiver demodulated each signal and converted the detected binary sequence back to text. The experiment demonstrates that higher modulation order can transmit more bits per symbol and reduce the required symbol rate, but more signal states must be distinguished at the receiver. As E_b/N_0 decreases, noise increases the probability of wrong decisions and BER rises until the message can no longer be reconstructed correctly. The final threshold values should be taken directly from the MATLAB observations. Among the tested schemes, QPSK offers a simple and useful compromise for a noisy channel, while 16-QAM provides a higher number

of bits per symbol when the channel condition is better.

10. References

- [1] American International University - Bangladesh (AIUB), Department of Computer Engineering, COE 3201 Data Communication Laboratory, Experiment 6 Student Manual, "Study of Digital to Analog Conversion using MATLAB."
- [2] American International University - Bangladesh (AIUB), Department of Computer Engineering, COE 3201 Data Communication Laboratory, Experiment 8 Student Manual, "Message Packing and Receiving Using Modulator."
- [3] American International University - Bangladesh (AIUB), Faculty of Engineering, Data Communication Lab, "Open-Ended Lab", Summer. 2025-26.